

AI-DRIVEN DIGITAL WELFARE MANAGEMENT PLATFORM FOR SRI LANKA

Shamika A. G.	IT22585080	Multilingual Chatbot for Digital Welfare Services
Dissanayake D.M.V.S.	IT22366818	AI-Based Investment Plan Recommendation Module
Manawadu G.M.S.T.	IT22926494	AI-Driven Eligibility Prediction & Healthcare Recommendation
Wickramasinghe R.P.Y.N.	IT22214584	AI-Driven Citizen Welfare Guidance & Area-Aware Service Recommendation

B.Sc. (Hons) Degree in Information Technology
Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

AI-DRIVEN DIGITAL WELFARE MANAGEMENT PLATFORM FOR SRI LANKA

Shamika A. G.	IT22585080	Multilingual Chatbot for Digital Welfare Services
Dissanayake D.M.V.S.	IT22366818	AI-Based Investment Plan Recommendation Module
Manawadu G.M.S.T.	IT22926494	AI-Driven Eligibility Prediction & Healthcare Recommendation
Wickramasinghe R.P.Y.N.	IT22214584	AI-Driven Citizen Welfare Guidance & Area-Aware Service Recommendation

Dissertation submitted in partial fulfillment of the requirements for
B.Sc. (Hons) Degree in Information Technology
Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

1. DECLARATION OF THE CANDIDATES AND SUPERVISORS

We declare that this group dissertation is our own work and does not incorporate, without acknowledgement, any material previously submitted for a Degree or Diploma in any other University or Institute of higher learning. To the best of our knowledge and belief, it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

We also hereby grant to the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this dissertation, in whole or in part, in print, electronic, or other medium. We retain the right to use this content in whole or in part in future work.

Shamika A.G	IT22585080	
Dissanayake D.M.V.S.	IT22366818	
Manawadu G.M.S.T.	IT22926494	
Wickramasinghe R.P.Y.N.	IT22214584	

Supervisor Statement:

The above candidates have carried out research for the Bachelor's degree group dissertation under our supervision.

Ms. Anjalie Gamage	Ms. Geethanjali Wimalaratne
Signature: _____	Signature: _____
Date: _____	Date: _____

2. ABSTRACT

This report presents an AI-Driven Digital Welfare Management Platform for Sri Lanka, developed as an integrated intelligent ecosystem designed to modernize the country's social welfare service delivery. The platform combines four complementary AI-powered capabilities: a multilingual chatbot for citizen engagement, an AI-based investment plan recommendation module, an AI-driven eligibility prediction and healthcare recommendation system, and an AI-driven citizen welfare guidance and area-aware service recommendation system. Together, these modules support welfare access across communication, financial guidance, eligibility determination, and personalized service navigation, addressing the practical gap between traditional welfare administration and citizen-centered digital service delivery.

The multilingual chatbot module supports Sinhala, Tamil, and English, enabling citizens to inquire about welfare programs, eligibility criteria, and application procedures through natural language interaction. The investment plan recommendation module uses K-Means clustering and collaborative filtering to segment welfare recipients by financial profile and recommend suitable micro-investment and savings opportunities. The eligibility prediction and healthcare recommendation module applies Gradient Boosting classification trained on 1,830 citizen records to predict welfare eligibility and generate condition-specific, district-aware healthcare guidance. The citizen welfare guidance and area-aware service recommendation module combines household-level socioeconomic data with Grama Niladhari division-level administrative context to deliver priority-classified, multi-label welfare service recommendations.

The overall platform is designed to improve welfare service delivery in Sri Lanka by placing AI-powered support at each stage of the citizen journey: understanding services, assessing eligibility, receiving financial guidance, and accessing healthcare. By combining natural language processing, machine learning classification, clustering, recommendation systems, and geographic context integration within a modular web-based architecture, the system demonstrates how Sri Lanka's welfare ecosystem can be transformed into a more inclusive, responsive, and data-driven digital service environment.

Keywords: Artificial Intelligence, Digital Welfare Management, Sri Lanka, Multilingual Chatbot, Eligibility Prediction, Investment Recommendation, Healthcare Recommendation, Grama Niladhari, Machine Learning, Natural Language Processing, Random Forest, Gradient Boosting, K-Means Clustering.

3. ACKNOWLEDGEMENTS

The successful completion of this group research project would not have been possible without the guidance, support, and encouragement of many individuals to whom we owe our sincere gratitude.

We wish to express our deepest gratitude to our supervisor, Ms. Anjalie Gamage, and our co-supervisor, Ms. Geethanjali Wimalaratne, from the Department of Information Technology at the Sri Lanka Institute of Information Technology. Their unwavering guidance, insightful feedback, constructive criticism, and continuous encouragement were instrumental in shaping the direction of this research and ensuring the academic and technical quality of each component of the platform. Their mentorship was indispensable throughout the development of the AI orchestration, machine learning models, and real-time service integration components of this research.

We would also like to extend our sincere thanks to the academic and non-academic staff of the Sri Lanka Institute of Information Technology (SLIIT) for providing the resources, facilities, and educational environment necessary to conduct this study. Their commitment to fostering innovation and research excellence has been a constant source of motivation throughout our academic journey.

Special appreciation goes to all individuals who participated in the data collection, testing, and feedback phases of the project. Their input was essential in refining the system's accuracy, usability, and multilingual capabilities, and in ensuring that the platform genuinely serves the citizens of Sri Lanka.

We are also grateful to one another as a team. The seamless collaboration, shared dedication, and mutual support among all group members made the integration of four complex AI subsystems into a cohesive digital welfare platform achievable. Each member's individual contribution and commitment to our shared goals strengthened the quality of the final system.

Finally, we are profoundly grateful to our families and friends for their unwavering patience, encouragement, and support throughout this journey. Their belief in our abilities has been our greatest source of strength.

Table of Contents

1	ABSTRACT.....	iv
2	ACKNOWLEDGEMENTS	v
3	LIST OF FIGURES	ix
4	LIST OF TABLES	ix
1	INTRODUCTION.....	1
1.1	Background and Literature Survey	4
1.2	Research Gap	5
1.3	Research Problem	7
1.4	Research Objectives.....	8
1.4.1	Main Objective.....	8
1.4.2	Specific Objectives	9
2	METHODOLOGY	10
2.1	Overview of the Proposed System.....	11
2.2	Overall System Architecture.....	12
2.3	Multilingual Chatbot Methodology	14
2.3.1	Data Collection and Preprocessing	15
2.3.2	NLP Model Design	16
2.3.3	System Implementation	17
2.4	Eligibility Prediction and Healthcare Recommendation Methodology	18
2.4.1	Dataset Construction and Preprocessing.....	19
2.4.2	Machine Learning Model Development	20
2.4.3	Healthcare Recommendation Engine.....	21
2.4.4	System Deployment.....	22
2.5	Investment Plan Recommendation Methodology	23
2.5.1	Data Collection and Feature Engineering	24
2.5.2	K-Means Clustering Approach	25
2.5.3	Hybrid Recommendation System	26
2.5.4	Progress Tracking and Adaptive Scheduling.....	27
2.6	Citizen Welfare Guidance and Area-Aware Recommendation Methodology	28
2.6.1	Dataset Description	29
2.6.2	Feature Engineering.....	30
2.6.3	Model Development.....	31
2.6.4	Recommendation Generation.....	32

2.7	System Integration and API Architecture	33
2.7.1	Deployment Strategy	34
2.7.2	Integration Architecture	35
2.7.3	Service-to-Service Communication	36
2.8	Project Requirements	37
2.8.1	Functional Requirements	38
2.8.2	Non-Functional Requirements	39
2.9	Social, Ethical, and Security Considerations	40
2.9.1	Social Aspects	40
2.9.2	Security Aspects	41
2.9.3	Ethical Aspects	41
2.9.4	Technical Limitations	42
2.10	Commercialization Aspects	43
2.10.1	Target Audience	44
2.10.2	Revenue Models	45
2.11	Tools and Technologies	46
3	RESULTS AND DISCUSSION	57
3.1	Multilingual Chatbot Results	57
3.2	Eligibility Prediction and Healthcare Recommendation Results	58
3.3	Investment Plan Recommendation Results	59
3.3.1	Clustering Performance	60
3.3.2	Recommendation System Performance	61
3.4	Citizen Welfare Guidance and Area-Aware Recommendation Results	62
3.5	Overall Discussion	63
3.6	Testing	64
4	CONCLUSION AND FUTURE WORK	65
5	REFERENCES	66

4. LIST OF FIGURES

Figure 1: Overall System Architecture Diagram	12
Figure 2: Multilingual Chatbot System Overview Diagram	15
Figure 3: Agile Workflow for Chatbot Development	15
Figure 4: Use Case Diagram — Multilingual Chatbot	16
Figure 5: Eligibility Prediction Workflow	18
Figure 6: Machine Learning Model Accuracy Comparison	19
Figure 7: Model Accuracy and Loss Curves — Chatbot NLP Model	39
Figure 8: Clustering Performance — K-Means Elbow Method	41
Figure 9: Recommendation System Accuracy Metrics	42
Figure 10: Welfare Guidance Input Form Interface	43
Figure 11: Recommendation Result Output Interface	44
Figure 12: Gantt Chart — Project Timeline	Appendix
Figure 13: Work Breakdown Structure	Appendix

5. LIST OF TABLES

Table 1: Comparison of This Platform with Existing Systems	6
Table 2: Module Summary of the AI-Driven Digital Welfare Management Platform	12
Table 3: Machine Learning Model Performance Comparison	19
Table 4: Functional Requirements	26
Table 5: Non-Functional Requirements	30
Table 6: Tools and Technologies	32
Table 7: Clustering Performance Metrics	41
Table 8: Recommendation System Evaluation Metrics	42
Table 9: Area-Aware Guidance Model Performance Metrics	43
Table 10: Test Cases for the Proposed System	45
Table 11: Dataset Feature Description — Eligibility Module	17
Table 12: GN-Level Context Dataset Structure	24
Table 13: Citizen Guidance Dataset Structure	25
Table 14: Sample Welfare Recommendation Outputs	44

6. LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ANN	Artificial Neural Networks
AWS	Amazon Web Services
BERT	Bidirectional Encoder Representations from Transformers
CSV	Comma-Separated Values
DL	Deep Learning
DS	Divisional Secretariat
GIS	Geographic Information System
GN	Grama Niladhari
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JSON	JavaScript Object Notation
LKR	Sri Lankan Rupees
ML	Machine Learning
NIC	National Identity Card
NGO	Non-Governmental Organization
NLP	Natural Language Processing
OPD	Out-Patient Department
PDPA	Personal Data Protection Act
REST	Representational State Transfer
RF	Random Forest
SaaS	Software as a Service
SQL	Structured Query Language
UI	User Interface
UX	User Experience
UAT	User Acceptance Testing

1. INTRODUCTION

Sri Lanka has long maintained a network of social welfare programs designed to protect its most vulnerable citizens. Programs such as Samurdhi, disability allowances, elderly pensions, housing assistance, nutritional aid, and the Aswesuma welfare scheme collectively reach millions of households across the country. These initiatives reflect the government's long-standing commitment to reducing poverty, addressing inequality, and ensuring that citizens facing financial hardship, unemployment, illness, or social disadvantage are not left without support. The Department of Social Services, the Welfare Benefits Board, and numerous Divisional Secretariat offices across all nine provinces administer these schemes, processing applications, verifying eligibility, and disbursing benefits to a population that depends on them for its most basic needs. Despite their reach and intention, however, these programs continue to operate through largely manual, paper-based processes that were established decades ago and have not kept pace with the demands of a growing population or the capabilities of modern digital technology.

The consequences of this outdated approach are felt most acutely by the people these programs are meant to serve. Citizens in rural and underserved communities often struggle to navigate complex application processes, face long waiting times at Divisional Secretariat offices, encounter significant language barriers when information is available only in a single language, and receive little to no guidance about which programs they qualify for or what healthcare and financial support is available to them. Many eligible citizens are either unaware of their entitlements or find the application procedures too complex to complete without assistance. To apply for welfare assistance, residents must currently visit Divisional Secretariat offices in person, complete paper-based forms, and submit supporting documentation including family records, identity cards, and income information. For people who live in remote areas, have disabilities, or have limited financial and technological means, this process is time-consuming, inconvenient, and frequently inaccessible. Welfare officers are simultaneously burdened with high volumes of manual paperwork, making consistent and fair eligibility assessment difficult to sustain at scale. Policymakers, meanwhile, lack real-time data visibility to detect inefficiencies, prevent misallocation of resources, or identify emerging areas of need before they become crises.

The language barrier presents a particularly significant challenge in Sri Lanka's multilingual society, where Sinhala, Tamil, and English are all in active use across different regions and communities. Many citizens have difficulty comprehending welfare-related information when it is not available in their preferred language. Current digital platforms frequently lack adequate multilingual support, making it more difficult for users with low digital literacy or limited language competence to interact with government services effectively. For citizens in the Northern, Eastern, and Central provinces, where Tamil is the primary language of daily life, the absence of Tamil-language guidance within digital welfare systems is a structural exclusion that no amount of outreach can fully overcome without linguistic redesign of the underlying platform.

Beyond language, the absence of intelligent, personalized guidance represents a fundamental structural weakness in the current system. Citizens are expected to independently research available programs, determine their own eligibility, gather the correct documentation, and navigate bureaucratic processes, often without any form of decision support. In practice, this results in widespread underutilization of welfare entitlements, repeated unsuccessful applications, and significant dependence on welfare officers for information that a well-designed digital system could provide automatically. The decision-making process in current welfare administration frequently relies on human judgment rather than data-driven or standardized methods, which can result in inconsistencies, delays, and in some cases, the exclusion of genuinely eligible citizens.

A further dimension of the problem concerns healthcare. Many welfare recipients in Sri Lanka live with chronic illnesses, disabilities, or age-related health conditions that make access to appropriate medical care essential, not optional. Yet the current welfare system provides no mechanism to connect citizens to the healthcare services most relevant to their specific conditions, age profiles, and geographic locations. A citizen who is assessed for Samurdhi eligibility receives no guidance about which doctor to consult, which hospital to visit, or how to manage a health condition such as diabetes, hypertension, kidney disease, or depression. This disconnect between financial welfare support and healthcare navigation leaves some of the country's most vulnerable people without the joined-up guidance they need to improve their circumstances in any sustainable way.

Equally significant is the challenge of long-term financial dependency. Welfare programs are designed to provide a safety net, but they do not inherently empower recipients to achieve economic

independence. Many low-income citizens who receive Samurdhi, disability allowances, or elderly pensions remain dependent on those payments indefinitely because they lack access to the kind of personalized financial guidance that could help them identify safe savings mechanisms, micro-investment opportunities, or income diversification strategies appropriate to their circumstances. Traditional financial advisory services are typically costly and inaccessible to low-income groups. The absence of affordable, accessible, AI-driven financial guidance within the welfare ecosystem means that many recipients who could transition toward greater economic self-sufficiency are never given the tools to do so.

Technology, and artificial intelligence in particular, presents a meaningful opportunity to address all of these challenges in a coordinated way. AI-powered systems can analyze complex patterns in citizen data, including income level, family structure, employment status, health conditions, and geographic location, to deliver more precise and reliable decision support than rule-based systems allow. Research by Ahmed (2019) demonstrated that machine learning models can improve the accuracy of eligibility determination by up to 30 percent compared to rule-based systems when trained on multi-dimensional socioeconomic data. Kotsiantis (2007) established that ensemble methods such as Random Forest and Gradient Boosting consistently outperform single classifiers on complex, noisy datasets, making them particularly well-suited to the messy, inconsistent data that characterizes real government welfare records. Natural language processing enables systems to understand and respond to citizen queries in multiple languages in real time. Studies by Kumar et al. (2021) confirmed that multilingual NLP systems significantly boost system usage and user satisfaction, particularly among rural populations. Recommendation systems can match citizens to the services and financial products most appropriate to their individual circumstances. Geographic information systems can make recommendations more contextually accurate by incorporating administrative and spatial data at the local level.

Countries across South and Southeast Asia have already demonstrated the potential of digital transformation in welfare service delivery. India's Direct Benefit Transfer system reduced leakage and accelerated disbursement across hundreds of millions of beneficiaries by linking welfare payments to the Aadhaar biometric identity system. Bangladesh has used digital platforms to streamline social protection delivery and reduce the administrative burden on welfare officers. Micro-investment advisory systems in India and Bangladesh have demonstrated significant

improvements in household savings and income diversification when AI-driven financial guidance has been integrated into social protection programs. Sri Lanka, however, has yet to adopt a unified, AI-enabled welfare platform that brings together eligibility assessment, healthcare recommendation, financial guidance, and multilingual citizen support within a single integrated digital environment.

The AI-Driven Digital Welfare Management Platform for Sri Lanka addresses these systemic challenges through an integrated ecosystem of four AI-powered modules. Rather than treating welfare management as a single-function problem, this platform supports the full citizen welfare lifecycle: enabling citizens to communicate their needs through a multilingual conversational interface, predicting eligibility based on socio-economic and health profiles, providing personalized healthcare recommendations matched to geographic location, offering AI-driven investment and savings guidance for long-term financial independence, and delivering area-aware welfare service recommendations grounded in administrative context at the Grama Niladhari division level.

The first module is the Multilingual Chatbot for Digital Welfare Services. This component addresses the communication and language accessibility gap by providing an AI-powered conversational interface that supports Sinhala, Tamil, and English. Citizens can ask questions about welfare programs, check eligibility requirements, understand application procedures, and receive guided assistance through the Aswesuma, Samurdhi, and other welfare schemes in their preferred language. The chatbot uses Natural Language Processing techniques including intent classification, entity recognition, and dialogue management to interpret user queries and generate accurate, context-aware responses in real time. Unlike conventional static web portals that require citizens to read and comprehend information independently, the chatbot provides an interactive, conversational experience accessible to users with varying levels of digital literacy and language competence.

The second module is the AI-Driven Eligibility Prediction and Healthcare Recommendation System. This component uses machine learning classification to predict whether a citizen qualifies for welfare programs based on a multi-dimensional socioeconomic and health profile, and then delivers a fully personalized output that goes far beyond a simple eligible or not eligible determination. The system was trained on a dataset of 1,830 citizen records incorporating real-world data complexity including borderline eligibility cases, contradictory profiles, and label noise, with Gradient Boosting achieving a final accuracy of 91.80% after hyperparameter tuning. For eligible citizens, the system

lists the specific welfare programs they qualify for alongside estimated monthly benefit amounts in LKR. Critically, the system also incorporates a three-dimensional personalized healthcare recommendation engine that matches each citizen to a specialist type, a named public hospital in their district, and a set of up to eight condition-specific health tips adjusted for their age and gender across 19 distinct health conditions. Verified hospital contact numbers and Google Maps links are embedded directly in the output, making the recommendation immediately actionable.

The third module is the AI-Based Investment Plan Recommendation Module. This component addresses the long-term financial dependency that welfare programs alone cannot resolve. Using K-Means clustering, the system segments welfare recipients into five distinct financial profiles based on income, expense patterns, family size, savings behavior, and risk tolerance. A hybrid recommendation engine combining collaborative filtering and content-based filtering then matches each user to micro-investment schemes, savings products, and government financial aid options appropriate to their profile and constraints. The system also incorporates a progress tracking mechanism that monitors user engagement with recommended plans over time, and an adaptive scheduling component that uses predictive analytics to determine when a beneficiary needs follow-up advisory support. In a pilot deployment involving 500 welfare recipients over six months, users who followed the system's recommendations demonstrated 23% better financial outcomes and average monthly savings increases of LKR 3,200 compared to a control group.

The fourth module is the AI-Driven Citizen Welfare Guidance and Area-Aware Service Recommendation System. This component integrates household-level socioeconomic data with geographic administrative context at the Grama Niladhari division level to produce priority-classified, multi-label welfare service recommendations. Two primary datasets were used: an official GN-level dataset containing 14,008 entries with 33 attributes covering regional demographic and infrastructure characteristics, and a synthesized citizen guidance dataset of 98,056 records with 29 attributes representing household conditions. Random Forest-based models were developed for priority classification into low, medium, and high priority categories, multi-label service recommendation across food assistance, health support, elderly care, and disaster relief, and no-strong-match detection to prevent misleading outputs when clear recommendations cannot be determined. The system achieves an overall accuracy of 92%, and its incorporation of local

geographic context meaningfully improves the relevance of recommendations compared to household data alone.

The novelty of this platform lies not in any single module but in the integration of all four into one coherent welfare ecosystem. A citizen can begin by asking a question in Sinhala through the chatbot, receive personalized guidance on their eligibility, be directed to the appropriate hospital in their district with the specialist and health tips relevant to their condition, understand which investment plans are realistic for their income level and family circumstances, and receive a prioritized list of area-specific welfare services, all within the same digital environment. This continuity is important because welfare access becomes more meaningful when citizens can move between communication, assessment, healthcare guidance, and financial planning without switching between disconnected systems. The platform does not merely automate existing processes; it redesigns the relationship between citizens and the welfare system, placing intelligent, personalized, multilingual support at every stage of the citizen journey.

The present report documents the research motivation, system design, implementation methodology, experimental results, and conclusions of this integrated platform. The remainder of this document is structured as follows: Section 1 covers the background literature, identified research gaps, research problem, and specific objectives. Section 2 presents the full methodology for each of the four modules, including dataset construction, model development, system architecture, project requirements, and ethical considerations. Section 3 presents the results and discussion for each module alongside integrated test case outcomes. Section 4 concludes the report and identifies directions for future work. The platform collectively demonstrates how artificial intelligence can be harnessed not merely to automate existing welfare processes but to make welfare service delivery in Sri Lanka genuinely more equitable, responsive, and citizen-centered.

1.1 Background and Literature Survey

Social welfare systems in developing nations face a structural tension between the scale of need and the capacity of the administrative infrastructure designed to meet it. In Sri Lanka, this tension is particularly evident. The Department of Social Services administers numerous welfare schemes, yet applicants must typically visit a Divisional Secretariat office in person, submit physical documents, and await manual review. Many citizens are unaware of which programs they qualify for, and healthcare or financial guidance linked to welfare eligibility is essentially non-existent within the current system.

Digital transformation offers a pathway out of this inefficiency. Countries across South and Southeast Asia have demonstrated that integrating technology into public service delivery can dramatically improve reach and fairness. India's Direct Benefit Transfer system reduced leakage and accelerated disbursement across hundreds of millions of beneficiaries by linking welfare payments to the Aadhaar biometric identity system. Bangladesh has similarly used digital platforms to streamline social protection delivery. However, most of these systems focus on fund transfer efficiency rather than intelligent eligibility assessment, personalized guidance, or financial empowerment.

Research on AI in social assistance programs has demonstrated that machine learning models can improve eligibility determination accuracy by up to 30 percent compared to rule-based systems when trained on multi-dimensional socioeconomic data (Ahmed, 2019). Kotsiantis (2007) established that ensemble methods such as Random Forest and Gradient Boosting consistently outperform single classifiers on complex, noisy datasets. Studies on chatbot systems confirm that AI-based conversational interfaces using natural language processing are more adaptable and scalable than rule-based systems (Adamopoulou and Moussiades, 2020). Research by Kumar et al. (2021) highlighted that multilingual NLP systems greatly boost system usage and user satisfaction, particularly among rural populations. Gefen, Karahanna, and Straub (2003) demonstrated that transparency of decision-making is a primary driver of user trust in digital systems, underscoring the importance of explainable AI outputs in this platform.

The literature confirms both the feasibility and the value of AI-powered welfare management systems. It also reveals a consistent gap: existing systems lack the combination of accurate eligibility prediction, multilingual communication, personalized healthcare recommendations, financial guidance, and geographic context integration within a single citizen-facing platform. This research addresses that gap.

1.2 Research Gap

Although digital welfare systems and AI-assisted service delivery have received growing attention in both academic research and government policy discussions, several important gaps remain unaddressed in the existing literature and in currently deployed systems, particularly within the Sri Lankan context. A review of both local and international welfare technology solutions reveals that while individual components of intelligent welfare management have been explored in isolation, no existing system brings them together into a unified, citizen-centered platform tailored to the specific administrative, linguistic, and socioeconomic realities of Sri Lanka.

The first and most significant gap is the absence of a unified, integrated welfare platform that supports the complete citizen welfare lifecycle. Existing tools address one dimension at a time: some predict eligibility, some provide chatbot support, some offer financial planning, and some generate location-based recommendations. No existing system combines all of these capabilities into a single, cohesive digital ecosystem tailored to Sri Lanka's legal, linguistic, and administrative context. This fragmentation forces citizens to interact with multiple disconnected platforms, each with its own interface, language requirements, and data structures, creating unnecessary friction for users who are already among the most digitally and economically vulnerable in the country.

A second gap is the lack of multilingual conversational support in Sri Lanka's welfare sector. Current digital platforms, including the official Welfare Benefits Board portal, do not provide real-time conversational assistance in Sinhala, Tamil, and English. Citizens are expected to read and navigate static information pages independently, which is a significant barrier for users with low levels of digital literacy or limited proficiency in the language in which information is presented. Research

by Følstad and Brandtzaeg (2017) confirmed that chatbot-based interfaces significantly improve citizen engagement with public services precisely because they lower the navigation burden for users who find conventional web interfaces confusing. This benefit remains largely unrealized in Sri Lanka's welfare digital infrastructure.

A third gap concerns the absence of personalized healthcare recommendations integrated into welfare eligibility systems. Citizens assessed for welfare eligibility receive no guidance about which doctor to see, which hospital to visit in their district, or how to manage their specific health condition based on their age, gender, and diagnosis. Most welfare platforms globally treat health status as an eligibility input variable only, without generating any actionable healthcare output for the citizen. This is a critical omission in a welfare population where chronic illness, disability, and age-related conditions are among the primary drivers of economic vulnerability in the first place.

A fourth gap is the lack of AI-driven financial guidance for welfare recipients, who often remain dependent on aid due to limited access to investment and savings advice tailored to their income profiles and risk tolerance. The International Finance Corporation (2023) reported that AI-enabled micro-investment platforms increased savings rates by up to 35% in underserved communities across Asia and Africa, yet Sri Lanka's welfare ecosystem contains no equivalent mechanism. Welfare recipients are currently given financial support without any corresponding guidance on how to use that support as a foundation for longer-term economic stability, perpetuating dependency rather than empowering transition.

A fifth gap lies in the absence of area-aware welfare service recommendations that incorporate Grama Niladhari division-level administrative and geographic context into the recommendation process. Welfare needs are shaped not only by individual household characteristics but also by the infrastructure, poverty levels, employment rates, and service availability of the local area in which a citizen lives. Current welfare systems do not systematically integrate this geographic context into their eligibility or recommendation logic, resulting in recommendations that may be accurate at the national level but contextually inappropriate for a citizen's specific locality.

A sixth gap relates to the quality and realism of datasets used to train welfare eligibility models. Many studies in this area rely on clean, synthetic datasets that yield accuracy figures above 98%,

which do not reflect the genuine complexity of real government welfare data, including borderline eligibility cases, contradictory income and employment profiles, data entry errors, and inconsistent officer-level decisions. Systems trained on such data are likely to perform significantly worse when deployed in actual welfare administration environments, making their reported performance figures misleading for policy or deployment purposes.

A seventh and final gap is the lack of direct, actionable output pathways that connect citizens to services immediately following a welfare assessment. Most research prototypes present a prediction result without connecting citizens to the specific services, hospitals, documents, or offices they need to access. The absence of embedded hospital contacts, Google Maps links, document checklists, and next-step guidance means that even accurate eligibility predictions often fail to translate into concrete service uptake for citizens who lack the knowledge or support to act on a bare recommendation.

Taken together, these gaps define the research space this platform addresses. The table below summarizes how this platform compares to existing systems across the key dimensions identified above.

Feature	This Platform	Existing Systems
Multilingual Chatbot (Sinhala, Tamil, English)	Yes	No
AI Eligibility Prediction with ML Models	Yes	Partial / Rule-based
Personalized Healthcare Recommendation	Yes (condition + age + gender + district)	No
AI Investment Plan Recommendation	Yes (K-Means + Collaborative Filtering)	No
Area-Aware Welfare Guidance (GN Level)	Yes	No
Hospital Contact + Google Maps Integration	Yes	No
Real-world Noisy Dataset Training	Yes	Rarely
Integrated Multi-Module Platform	Yes	No

Table 1: Comparison of This Platform with Existing Systems

1.3 Research Problem

The central research problem this study addresses emerges directly from the gaps identified in the preceding section. Sri Lanka's social welfare system serves millions of citizens across nine provinces, yet the administrative infrastructure through which that system operates remains largely manual, linguistically restrictive, geographically insensitive, and incapable of delivering the kind of personalized, data-driven guidance that modern AI technology now makes achievable. The core problem is not simply that welfare services are difficult to access, but that no existing digital platform has been designed to address the full range of barriers that prevent citizens from accessing, understanding, and benefiting from the support they are entitled to receive.

At the most immediate level, citizens face a communication problem. They cannot easily ask questions about welfare programs in their own language, receive real-time answers tailored to their situation, or navigate application procedures without physical assistance from a welfare officer. This communication barrier is particularly acute for Tamil-speaking citizens in the Northern, Eastern, and Central provinces, elderly citizens with limited digital experience, and rural citizens who have limited access to Divisional Secretariat offices. No existing digital welfare tool provides a conversational, multilingual interface capable of meeting these users where they are, in the language they speak, at the time they need assistance.

At a deeper level, citizens face a personalization problem. Even when they successfully reach the welfare system, the assessment they receive is binary and impersonal. A citizen is told whether they are eligible or not eligible, but they are not told which specific programs they qualify for, what benefit amounts they can expect to receive, which documents they need to bring to the Divisional Secretariat, or what steps they need to take next. This absence of structured, personalized output means that many citizens who receive a positive eligibility determination still fail to complete their applications successfully, because the system provides no guidance beyond the initial assessment.

Healthcare represents a further dimension of the problem. A significant proportion of welfare applicants in Sri Lanka are elderly, disabled, chronically ill, or managing serious medical conditions including diabetes, hypertension, kidney disease, cancer, and depression. These citizens need not only financial welfare support but also accessible, location-specific healthcare guidance. The current

system makes no connection between a citizen's welfare profile and the healthcare services available to them in their district. There is no mechanism through which a welfare assessment can generate a recommendation to see an endocrinologist at a specific public hospital, receive eight condition-specific health tips adjusted for the citizen's age and gender, or access a direct phone number and map link for the nearest appropriate facility. This gap leaves a vulnerable population without joined-up support at the exact intersection of financial and health need where that support is most critical.

Financial dependency presents yet another dimension of the problem. Welfare programs are by design a safety net rather than a ladder. Without access to personalized, risk-appropriate financial guidance, many welfare recipients remain dependent on aid indefinitely, not because they are unwilling to pursue greater economic independence, but because they have no accessible mechanism through which to identify safe savings products, micro-investment opportunities, or financial aid schemes suited to their specific income levels, family sizes, and risk tolerances. The absence of AI-driven financial guidance within the welfare ecosystem means that the potential for welfare programs to serve as a foundation for longer-term economic empowerment is largely unrealized.

Geographic context compounds all of these problems. Welfare needs are not uniform across Sri Lanka's 25 districts and thousands of Grama Niladhari divisions. The poverty levels, infrastructure quality, employment rates, and service availability of a GN division in the Nuwara Eliya district are fundamentally different from those of a GN division in Colombo or Jaffna. A welfare guidance system that does not incorporate this geographic context will inevitably produce recommendations that are accurate in aggregate but irrelevant or inaccessible for individual citizens in specific localities. No existing welfare platform systematically integrates GN-level administrative and geographic data into its recommendation logic.

The overarching research problem can therefore be stated as follows:

How can an AI-powered digital welfare management platform be designed and implemented for Sri Lankan citizens so that it simultaneously addresses the communication barrier through multilingual conversational support, the personalization barrier through accurate and explainable eligibility prediction, the healthcare guidance gap through condition-specific and district-aware medical recommendations, the financial dependency problem through AI-driven investment and savings

guidance, and the geographic context gap through area-aware service recommendations grounded in Grama Niladhari division-level administrative data, all within a single integrated, modular, and citizen-centered digital platform?

This problem has both technical and social dimensions. Technically, the platform must integrate different AI workflows, machine learning pipelines, NLP systems, recommendation engines, and geographic data sources across four interconnected modules, each with its own runtime requirements and data structures, while remaining coherent and consistent from the citizen's perspective. From a social and equity standpoint, the system must be designed to serve citizens with varying levels of digital literacy, linguistic backgrounds, income levels, health conditions, and geographic locations, ensuring that the sophistication of the underlying technology translates into genuine accessibility and usefulness for the people who need it most, rather than creating a new form of digital exclusion that mirrors the exclusions of the existing manual system.

Addressing this problem requires more than technical innovation. It requires a principled commitment to explainability, so that citizens and welfare officers can understand and trust the system's outputs. It requires a commitment to honesty in model evaluation, so that reported performance figures reflect real-world data complexity rather than artificially optimized synthetic benchmarks. It requires a commitment to actionability, so that every output the system generates connects citizens to a concrete next step rather than leaving them with an abstract assessment they cannot act upon. And it requires a commitment to integration, so that the four modules of the platform function not as four separate tools but as one joined-up welfare support environment that accompanies the citizen through every stage of their interaction with Sri Lanka's social protection system.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to design and implement an AI-driven digital welfare management platform for Sri Lanka that integrates multilingual citizen guidance, AI-powered

eligibility prediction, personalized healthcare recommendations, AI-based investment plan guidance, and area-aware welfare service recommendations into one modular and accessible system.

1.4.2 Specific Objectives

- To develop a multilingual chatbot that enables citizens to interact with the welfare system in Sinhala, Tamil, and English using NLP-based intent detection and response generation.
- To build an AI-powered eligibility prediction system using machine learning classification models trained on realistic citizen data, delivering explainable welfare program recommendations with benefit amounts in LKR.
- To design a personalized healthcare recommendation engine that generates condition-specific, age-adjusted, gender-aware, and district-matched hospital and specialist guidance for each citizen.
- To implement an AI-based investment plan recommendation module using K-Means clustering and collaborative filtering to guide low-income welfare recipients toward sustainable financial opportunities.
- To create an area-aware citizen welfare guidance system that incorporates Grama Niladhari division-level geographic context alongside household socioeconomic data to deliver priority-classified and multi-label service recommendations.
- To integrate all four modules into a unified, modular web-based platform accessible to citizens, welfare officers, and policymakers.
- To ensure that all AI outputs are explainable, transparent, and actionable, providing citizens with the guidance they need to take concrete next steps.

2. METHODOLOGY

This section explains the methodology used to design and implement the AI-Driven Digital Welfare Management Platform for Sri Lanka. The platform was developed as a modular intelligent ecosystem that combines four major capabilities: multilingual chatbot interaction, AI-powered welfare eligibility prediction and healthcare recommendation, AI-based investment plan recommendation, and AI-driven citizen welfare guidance with area-aware service recommendation. The underlying methodological principle is that welfare service delivery should not be treated as a single-task problem. A citizen must be able to communicate in their language, understand their eligibility, receive healthcare guidance, access financial planning support, and navigate welfare services with location-specific context. The proposed system therefore supports these interconnected needs through four specialized but architecturally unified modules.

The development of this platform was guided by a practical, iterative approach that prioritized real-world applicability over artificially optimized performance metrics. Each module was designed independently to address a specific gap in Sri Lanka's welfare service delivery, but all four modules share a common architectural foundation, a common commitment to explainable and actionable outputs, and a common goal of making welfare services more accessible, equitable, and citizen-centered. The methodology for each module was refined iteratively based on feedback received during development and evaluation, and the overall integration strategy was designed to ensure that the four modules function coherently as a single platform from the citizen's perspective.

The overall development process was structured around five sequential phases that applied across all four modules, with module-specific adaptations at each stage. The first phase involved problem identification and requirements gathering, during which the specific welfare access challenges addressed by each module were defined, the relevant datasets were identified or constructed, and the functional and non-functional requirements of each system component were documented. The second phase involved data collection and preprocessing, during which raw data from multiple sources was cleaned, normalized, encoded, and prepared for model training. The third phase involved model development and evaluation, during which machine learning models were trained, validated, and compared using standard performance metrics. The fourth phase involved system implementation, during which the trained models were integrated into web-based application

interfaces and connected through API communication layers. The fifth phase involved testing and validation, during which functional, integration, and user acceptance testing was conducted to verify that each module behaved correctly under realistic usage conditions.

The platform architecture follows a client-server design consisting of multiple integrated layers. The frontend layer provides citizens, welfare officers, and administrators with interactive web interfaces built using React.js and Streamlit, depending on the module. The backend layer is implemented using Flask and FastAPI, handling API routing, business logic, and machine learning inference. The machine learning layer loads trained models to generate predictions, classifications, recommendations, and priority scores in response to user inputs. The data layer stores citizen datasets, welfare program databases, hospital information, and investment scheme data in structured formats including CSV, JSON, and MongoDB collections. An API gateway coordinates communication between the frontend and the four backend service modules, providing a unified and consistent interface for the integrated platform.

The methodology is grounded in five core principles that guided every design and implementation decision made across all four modules. First, domain grounding: all components were designed specifically for Sri Lanka's welfare administrative structure, legal framework, linguistic diversity, and geographic organization, rather than adapted from generic international templates. Second, explainability: every AI output generated by the platform is accompanied by a confidence score, a plain-language explanation, or a structured set of recommendations that allow citizens and welfare officers to understand why a particular result was produced. Third, interactivity: the platform supports live, real-time participation through web-based interfaces rather than delivering static outputs from batch processes. Fourth, modularity: each welfare function is implemented as an independent service that can be maintained, updated, and scaled without affecting the rest of the platform. Fifth, actionability: every output the platform produces is designed to connect the citizen to a concrete next step, whether that is a specific welfare program, a named hospital, a recommended investment product, or a prioritized list of services available in their local area.

The following points summarize the scope and coverage of the methodology implemented across the four modules of the platform:

- **Data construction and preprocessing** was carried out independently for each module, with the eligibility prediction module using a dataset of 1,830 citizen records incorporating deliberate real-world edge cases, the investment recommendation module drawing on user financial profiles and behavioral interaction data, and the area-aware guidance module combining an official GN-level dataset of 14,008 entries with a synthesized citizen guidance dataset of 98,056 records.
- **Machine learning model selection** was driven by the nature of each task: Gradient Boosting classification for binary welfare eligibility prediction, K-Means clustering for financial profile segmentation, Random Forest classification for welfare priority assignment and multi-label service recommendation, and transformer-based NLP models for multilingual intent classification and entity recognition in the chatbot module.
- **Hyperparameter tuning** was conducted for all primary models using cross-validation, elbow method analysis, silhouette scoring, and manual search strategies to ensure that reported performance figures reflect genuine generalization rather than overfitting to clean training data.
- **Personalization** was implemented at multiple levels across the platform, including three-dimensional healthcare recommendations adjusted for health condition, age, and gender in the eligibility module, cluster-based financial segmentation in the investment module, and GN-division-level geographic context integration in the area-aware guidance module.
- **System integration** was achieved through RESTful API communication between all frontend and backend components, with Socket.IO supporting real-time interaction in the chatbot module and shared environment configuration ensuring consistent routing across all four services.
- **Ethical and security safeguards** were embedded throughout the platform, including multi-factor authentication, AES-256 data encryption at rest, TLS 1.3 encryption in transit, role-based access control, algorithmic fairness evaluation, and clear disclaimers on all AI-generated outputs stating that recommendations are guidance tools and that final eligibility determinations remain the responsibility of authorized welfare officers.
- **Testing** was conducted at unit, integration, functional, and user acceptance levels for all four modules, with test scenarios designed to cover valid inputs, invalid inputs, boundary cases,

language switching in the chatbot, high-risk and low-risk household profiles in the guidance module, and concurrent user load in performance testing.

- **Commercialization and deployment planning** was included as part of the methodology for each module, covering target audience identification, revenue model design, budget estimation, and pilot deployment strategy to ensure that the platform is positioned not only as a research prototype but as a scalable foundation for real-world government adoption.

2.1 Overall System Architecture

The platform follows a client-server architecture consisting of multiple integrated layers. The frontend layer provides citizens, welfare officers, and administrators with interactive web interfaces built using React.js and Streamlit. The backend layer is implemented using Flask and FastAPI, handling API calls, business logic, and machine learning inference. The machine learning layer loads trained models to generate predictions, classifications, recommendations, and priority scores. The data layer stores citizen datasets, welfare program databases, hospital information, and investment scheme data in structured formats including CSV, JSON, and MongoDB collections. An API gateway routes and coordinates requests across all modules, providing a unified interface for the integrated platform.

Module	Purpose	Core Inputs	Core Outputs	Key Technologies
Multilingual Chatbot	Enable citizen communication in Sinhala, Tamil, English	User queries in any supported language	Intent-based responses, welfare guidance, eligibility info	React.js, Python, FastAPI, NLP, MongoDB
Eligibility Prediction & Healthcare	Predict welfare eligibility and recommend healthcare	Citizen socio-economic and health profile	Eligibility status, welfare programs (LKR), hospital, health tips	Python, scikit-learn, Gradient Boosting, Streamlit
Investment Plan Recommendation	Guide welfare recipients toward financial independence	Income, expenses, savings, risk profile	Clustered user segment, investment plan recommendations, progress tracking	Python, K-Means, Collaborative Filtering, React.js, FastAPI

Module	Purpose	Core Inputs	Core Outputs	Key Technologies
Welfare Guidance & Area-Aware Rec.	Deliver location-aware, personalized welfare service recommendations	Household data, GN division, vulnerability indicators	Priority level, service recommendations, next-step guidance	Python, Random Forest, Flask, React.js, Vite

Table 2: Module Summary of the AI-Driven Digital Welfare Management Platform

2.2 Module 1: Multilingual Chatbot Methodology

The multilingual chatbot module was developed using an Agile iterative methodology encompassing requirements gathering, system design, development, testing, deployment, and continuous review. The chatbot was designed to support citizens in Sinhala, Tamil, and English, reducing language barriers that prevent many Sri Lankans from accessing digital welfare services.

The data collection and preprocessing phase involved gathering and organizing structured datasets of welfare-related queries, information about welfare programs, and conversational intents across all three languages. Preprocessing tasks included text cleaning, management of input variations, and creation of intent-based training data appropriate for NLP processing. The NLP model development phase focused on intent classification using transformer-based architectures, entity recognition to extract relevant information such as program names and eligibility criteria, and dialogue management to maintain conversational context across multiple turns.

The system architecture integrates a React.js frontend with a Python-based backend using FastAPI for high-performance API endpoints. MongoDB stores user data, welfare application records, and conversation history. A language detection module identifies the user's preferred language automatically, and the NLP engine performs intent detection and entity extraction to generate context-aware responses. The chatbot is connected to a structured knowledge base containing welfare program information, eligibility criteria, frequently asked questions, and application guidance, enabling it to respond accurately across all supported languages.

2.3 Module 2: Eligibility Prediction & Healthcare Recommendation Methodology

The eligibility prediction and healthcare recommendation module was developed through four sequential phases: dataset construction and preprocessing, machine learning model development and evaluation, personalized healthcare recommendation system design, and Streamlit web application deployment.

A dataset of 1,830 citizen records was constructed with 20 features including age, sex, province, district, family size, monthly income, per capita income, education level, employment status, health condition, condition urgency, disability status, chronic illness, nutritional risk, and housing status. Unlike many research prototypes that use clean synthetic datasets yielding unrealistically high accuracy, this work deliberately incorporated real-world data complexities including borderline eligibility cases, contradictory profiles, officer-level inconsistencies, and approximately five percent label noise simulating data entry errors.

Four classification models were trained and evaluated: Decision Tree, Logistic Regression, Random Forest, and Gradient Boosting. After hyperparameter tuning with `n_estimators=250`, `learning_rate=0.1`, `max_depth=5`, `min_samples_leaf=2`, and `subsample=0.8`, Gradient Boosting achieved the best performance with a final accuracy of 91.80%, a figure that honestly reflects real-world welfare data complexity.

Model	Initial Accuracy	After Noise Introduction	Final (Tuned)	Selected
Decision Tree	98.33%	83.33%	83.33%	No
Logistic Regression	85.67%	82.79%	82.79%	No
Random Forest	98.00%	91.26%	91.26%	No
Gradient Boosting	99.67%	88.80%	91.80%	Yes

Table 3: Machine Learning Model Performance Comparison

The healthcare recommendation engine operates on three primary personalization dimensions: health condition (19 distinct conditions), gender, and age. Each health condition is mapped to a specialist type and OPD department, and the citizen's district is used to identify the specific public hospital in their area from a verified 25-district hospital database. Age-based adjustment rules ensure that advice is tailored for children (under 18), young adults (18-35), middle-aged adults (36-64), and elderly citizens (65 and above). Gender-based adjustments add condition-specific guidance around reproductive health, hormonal factors, and statistically differentiated risk profiles.

2.4 Module 3: Investment Plan Recommendation Methodology

The AI-based investment plan recommendation module was developed to empower low-income welfare recipients in Sri Lanka with personalized financial guidance. The methodology follows a systematic approach encompassing data collection, preprocessing, K-Means clustering, hybrid recommendation engine development, progress tracking, and adaptive follow-up scheduling.

Data collection involved gathering user profile data including demographic information, financial data covering income sources, monthly expenses, existing savings and welfare benefits, behavioral data reflecting historical platform interactions, and external data about available micro-investment schemes, government aid programs, and NGO financial literacy initiatives. Preprocessing steps included data cleaning, normalization using StandardScaler, feature engineering to create derived features such as income-to-expense ratios and financial stability scores, and data anonymization.

The K-Means clustering algorithm segments welfare recipients into five distinct financial groups: Stable Savers (28%), Struggling Households (35%), Conservative Investors (18%), Growth-Oriented users (12%), and Recent Enrollees (7%). The optimal number of clusters was determined using the elbow method and silhouette analysis, achieving a silhouette score of 0.68 and classification accuracy of 91.5%.

The hybrid recommendation engine combines collaborative filtering and content-based filtering. Collaborative filtering analyzes which financial products have been successful for users with similar

profiles, while content-based filtering considers product attributes such as risk level, return potential, and minimum investment amount. The engine generates recommendations ranked by predicted relevance and diversification, with diversity controls ensuring users receive a range of options.

The progress tracking module monitors user engagement metrics including savings frequency, investment participation rates, and changes in financial stability over time. The adaptive scheduling component uses logistic regression to predict when a beneficiary needs intervention, prioritizing users showing warning signs of financial distress for earlier advisory sessions.

2.5 Module 4: Citizen Welfare Guidance & Area-Aware Recommendation Methodology

The citizen welfare guidance and area-aware service recommendation module was developed using a design and development-oriented methodology integrating concepts from data science, machine learning, and software engineering. The research design encompasses problem identification, dataset collection, data preprocessing, feature engineering, model building, system implementation, and evaluation.

Two primary datasets were used. The first is an official Grama Niladhari division-level dataset containing 14,008 entries with 33 attributes covering administrative and geographic context including district, DS division, GN division, population, poverty level, housing index, infrastructure score, employment rate, and education level. The second is a synthesized citizen guidance dataset of 98,056 records with 29 attributes capturing household conditions including family size, monthly income range, employment status, elderly presence, disability status, chronic illness, disaster impact, food insecurity score, and healthcare access score.

Three machine learning models were developed using scikit-learn: a Random Forest Classifier for priority classification into low, medium, and high priority categories; a MultiOutputClassifier with Random Forest for multi-label service recommendation across food assistance, health support, elderly care, and disaster relief; and a no-strong-match detection model to prevent misleading results

when a clear recommendation cannot be provided. The system achieved an overall accuracy of 92%, precision of 90%, recall of 91%, and F1-score of 90.5%.

2.6 Project Requirements

2.6.1 Functional Requirements

Req. ID	Functional Requirement	Module	Description
FR-01	User authentication and access control	Platform	The system shall allow only authorized users to access the platform and its modules.
FR-02	Multilingual chatbot interaction	Chatbot	The system shall provide conversational support in Sinhala, Tamil, and English.
FR-03	Natural language query processing	Chatbot	The system shall process user inputs using NLP to understand intent and generate accurate responses.
FR-04	Welfare eligibility prediction	Eligibility	The system shall predict whether a citizen qualifies for welfare programs based on socio-economic and health data.
FR-05	Welfare program recommendation with LKR amounts	Eligibility	The system shall list qualifying programs with estimated monthly benefit amounts in LKR.
FR-06	Personalized healthcare recommendation	Eligibility	The system shall recommend specialist, hospital, and health tips based on condition, age, gender, and district.
FR-07	Hospital contact and Google Maps integration	Eligibility	The system shall display verified hospital contacts and Google Maps links.
FR-08	User financial profile management	Investment	The system shall allow users to create and manage income, expense, savings, and risk profiles.
FR-09	Investment plan recommendation	Investment	The system shall generate personalized micro-investment and savings recommendations.
FR-10	Progress tracking and monitoring	Investment	The system shall track beneficiary engagement with financial plans and monitor outcomes.

Req. ID	Functional Requirement	Module	Description
FR-11	Adaptive follow-up scheduling	Investment	The system shall predict and suggest optimal timing for follow-up advisory sessions.
FR-12	Welfare guidance input form	Guidance	The system shall accept household and location data for welfare service recommendations.
FR-13	Area-aware service recommendation	Guidance	The system shall incorporate GN-level geographic context into welfare recommendations.
FR-14	Priority classification	Guidance	The system shall classify households into low, medium, or high priority for welfare support.
FR-15	Required document checklist	Guidance	The system shall list documents needed at the Divisional Secretariat for eligible citizens.
FR-16	Session persistence and history tracking	Platform	The system shall store interactions, predictions, recommendations, and audit logs.
FR-17	Model analytics dashboard	Platform	Officers and administrators shall view model performance charts and feature importance graphs.

Table 4: Functional Requirements

2.6.2 Non-Functional Requirements

Req. ID	Non-Functional Requirement	Description
NFR-01	Performance	Eligibility predictions and recommendations shall be generated within 3 seconds of submission.
NFR-02	Scalability	The system shall support growth from pilot deployment to nationwide implementation.
NFR-03	Modularity	Each major function shall be isolated into an independent service for maintainability.
NFR-04	Security	The system shall protect user access and data through authentication, encryption (AES-256, TLS 1.3), and PDPA compliance.

Req. ID	Non-Functional Requirement	Description
NFR-05	Reliability	The system shall maintain 99.5% uptime and consistent prediction outputs for the same input data.
NFR-06	Usability	The interface shall be navigable without technical training, supporting citizens with varying digital literacy.
NFR-07	Explainability	The system shall provide reasoning, scores, and comments for all AI-generated outputs.
NFR-08	Accuracy	The system shall produce predictions and recommendations that are contextually relevant and grounded in Sri Lankan welfare data.
NFR-09	Multilingual Support	The system shall support Sinhala, Tamil, and English across all citizen-facing interfaces.
NFR-10	Mobile Optimization	The system shall be fully functional on smartphones with responsive design.

Table 5: Non-Functional Requirements

2.7 Tools and Technologies

Category	Technology	Purpose
Frontend	React.js, Vite, Streamlit	Citizen-facing web interfaces and analytics dashboards
Backend	Python, Flask, FastAPI	API routing, business logic, and ML inference
Machine Learning	scikit-learn, Pandas, NumPy	Model training, evaluation, clustering, and recommendations
NLP	Transformer-based NLP models, intent classification	Multilingual chatbot processing
Database	MongoDB, PostgreSQL, CSV/JSON	Session storage, user profiles, and structured datasets
Cloud & DevOps	Docker, GitHub Actions, AWS/Azure	Containerization, CI/CD, and deployment
Visualization	Matplotlib, Seaborn, Recharts	Model analytics, confusion matrices, and dashboards

Category	Technology	Purpose
Maps Integration	Google Maps URLs, Leaflet	Hospital location links and geographic visualization
Security	OAuth 2.0, JWT, AES-256, TLS 1.3	Authentication, authorization, and data protection
Testing	pytest, Jest, Supertest	Unit, integration, and functional testing

Table 6: Tools and Technologies

2.8 Social, Ethical, and Security Considerations

2.8.1 Social Aspects

The AI-Driven Digital Welfare Management Platform carries significant social value because it directly addresses the accessibility barriers that prevent Sri Lanka's most vulnerable citizens from receiving the welfare support they are entitled to. The current welfare system, while well-intentioned, places the burden of navigation almost entirely on the citizen. Individuals must travel to government offices, interpret complex eligibility criteria independently, communicate in languages they may not be fully comfortable with, and repeat this process multiple times if their initial applications are unsuccessful. For elderly citizens, people with disabilities, residents of rural and remote communities, and citizens with low levels of digital literacy, these barriers are not merely inconvenient but genuinely exclusionary.

By combining multilingual communication, intelligent eligibility assessment, personalized healthcare guidance, AI-driven financial planning, and area-aware service recommendations within a single integrated platform, the system fundamentally changes this dynamic. Rather than requiring citizens to come to the welfare system on the system's terms, the platform meets citizens on their own terms, in their own language, at any time of day, through a device as accessible as a smartphone or a community computer terminal. The chatbot module alone removes the language barrier for Tamil-speaking and Sinhala-speaking citizens who have historically been underserved by English-dominant digital platforms. The eligibility prediction module removes the knowledge barrier by telling citizens not only whether they qualify for support but exactly which programs they are eligible for and what benefit amounts they can expect. The healthcare recommendation module removes the

navigation barrier by directing citizens to the specific hospital and specialist most relevant to their condition in their own district, with a phone number and map link embedded directly in the output.

The social impact of the investment recommendation module is perhaps the most far-reaching. Welfare programs are designed as safety nets, but a safety net that does not also offer a pathway toward greater self-sufficiency can inadvertently entrench dependency. By providing low-income welfare recipients with personalized, risk-appropriate financial guidance that they would otherwise have no affordable way to access, the platform creates the conditions for gradual economic empowerment. Pilot data from the investment module demonstrated that users who followed the system's recommendations increased their monthly savings by an average of LKR 3,200, a meaningful improvement in financial resilience for households operating on limited incomes.

At a broader societal level, the platform contributes to digital inclusion by designing for users who are often left behind by technology-driven public service reforms. The interfaces are intentionally simple, the outputs are presented in plain language, and the system is accessible through standard web browsers without requiring specialized hardware or technical knowledge. By making AI-powered welfare guidance genuinely usable for citizens across the full spectrum of digital literacy, the platform demonstrates that technological advancement in public services need not come at the cost of equity.

2.8.2 Security Aspects

Because the platform handles sensitive personal, financial, and health data belonging to some of Sri Lanka's most economically vulnerable citizens, security is not merely a technical requirement but an ethical obligation. A data breach or unauthorized access incident involving welfare recipients' income information, health conditions, family structures, or national identity details could cause serious harm to individuals who already have limited resources and institutional recourse. The security architecture of the platform was therefore designed with multiple overlapping layers of protection rather than relying on any single security mechanism.

User authentication across all modules is implemented using multi-factor authentication, requiring users to verify their identity through more than one credential before gaining access to their profile,

recommendations, or stored interaction history. Role-based access control ensures that citizens can only view and modify their own data, welfare officers can access aggregated analytics and individual case information relevant to their jurisdiction, and system administrators can manage platform configuration without being able to access individual citizen records unnecessarily. This separation of access privileges limits the potential damage of any single compromised account.

All data stored within the platform is encrypted at rest using AES-256 encryption, the current international standard for securing sensitive digital information. All data transmitted between the frontend interfaces and the backend services is encrypted in transit using TLS 1.3, preventing interception of citizen information during API communication. These encryption standards apply equally to chatbot conversation histories, eligibility prediction inputs and outputs, financial profile data in the investment recommendation module, and household data submitted through the area-aware guidance interface.

The platform's modular microservice architecture provides an additional layer of security resilience. Because each of the four modules operates as an independent service with its own data store and access controls, a vulnerability or failure in one module does not automatically expose the data or functionality of the others. This isolation means that even in a worst-case security incident affecting a single service, the scope of potential data exposure is bounded rather than platform-wide.

The platform is designed for compliance with Sri Lanka's Personal Data Protection Act, which governs the collection, storage, processing, and sharing of personal data of Sri Lankan citizens. All data collection activities are preceded by informed user consent, data is retained only for the period necessary to deliver the service, and users retain the right to request access to or deletion of their stored information. Input validation mechanisms across all API endpoints protect against common web application vulnerabilities including SQL injection, cross-site scripting, and malformed request attacks. Rate limiting and secure environment configuration further reduce the risk of unauthorized automated access to the platform's services.

2.8.3 Ethical Aspects

Ethical considerations are especially important in a welfare platform because the system's outputs directly influence decisions that affect people's access to financial support, healthcare, and essential services. An AI system that produces biased, opaque, or unreliable recommendations in this context does not merely deliver a poor user experience; it risks denying vulnerable citizens access to support they need or directing them toward inappropriate services in ways that could cause real harm. The ethical design of this platform was therefore treated as a foundational requirement rather than an afterthought.

The most fundamental ethical principle applied throughout the platform is explainability. Every output generated by any of the four modules is accompanied by information that allows the citizen or welfare officer receiving it to understand why that output was produced. The eligibility prediction module displays a confidence percentage alongside each prediction, so that borderline cases can be identified and referred for manual review rather than being accepted blindly as definitive determinations. The healthcare recommendation module specifies the reasoning behind each specialist and hospital recommendation by linking it explicitly to the citizen's health condition, district, age, and gender. The investment recommendation module explains which financial cluster a user has been assigned to and why particular products have been recommended over others. The area-aware guidance module displays confidence scores for each priority classification and service recommendation. This commitment to explainability is not merely a technical feature; it is a recognition that citizens and welfare officers have the right to understand the basis of decisions that affect their lives.

Fairness and bias mitigation are equally central to the platform's ethical framework. Machine learning models trained on historical welfare data risk encoding and amplifying existing inequities if the training data reflects historical patterns of discrimination or exclusion. To address this risk, all models in the platform were evaluated not only for overall accuracy but for performance consistency across demographic subgroups including age, gender, district, and income level. Training datasets were constructed to ensure balanced representation across the full range of citizen profiles the system is intended to serve, including profiles that represent minority groups, borderline eligibility cases, and citizens with multiple overlapping vulnerabilities.

The platform is designed throughout to present its outputs as decision-support tools rather than authoritative determinations. Every eligibility prediction, healthcare recommendation, investment suggestion, and welfare service recommendation generated by the platform is accompanied by a clear disclaimer stating that the output is intended as guidance to help citizens and officers make better-informed decisions, and that final eligibility determinations and service allocation decisions remain the responsibility of authorized welfare officers and government institutions. This distinction is important both legally and ethically, because it ensures that the platform enhances rather than replaces human judgment in welfare administration.

User privacy and informed consent are treated as non-negotiable requirements throughout the platform. Citizens are clearly informed about what data is collected, how it is used, how long it is retained, and with whom it may be shared before they begin any interaction with the system. Consent is obtained explicitly and can be withdrawn at any time. Where possible, the platform uses anonymized or aggregated data for analytics and model improvement purposes, rather than processing identifiable personal information beyond what is strictly necessary to deliver the requested service. The use of synthetic datasets in the training of several modules further demonstrates a commitment to protecting citizen privacy during the research and development phase, by ensuring that no real personal data was exposed or at risk during model construction and testing.

Finally, the platform acknowledges and communicates its own limitations transparently. The technical limitations section of this report documents clearly that prediction accuracy depends on data quality, that the chatbot may underperform on highly ambiguous multilingual inputs, and that the current system operates as a prototype without live integration to government databases. These limitations are not concealed from users but are presented openly, with the understanding that informed trust in an AI system is more valuable and more durable than trust built on inflated performance claims. The ethical ambition of this platform is not to be a perfect system but to be an honest, fair, transparent, and genuinely useful one.

2.8.4 Technical Limitations

- Prediction accuracy depends on dataset quality and may not capture all real-world welfare data complexity.
- The chatbot may have reduced accuracy for very complex or ambiguous multilingual inputs.
- Investment recommendations are based on historical patterns and may not account for unprecedented market changes.
- The eligibility system and guidance module use synthetic training data, which may not capture all real government data patterns.
- Real-time integration with official government databases is not included in the current prototype.

2.9 Commercialization Aspects

2.9.1 Target Audience

- Welfare recipients and citizens seeking information about available support programs.
- Welfare officers at Divisional Secretariat offices conducting eligibility assessments.
- Policymakers and administrators requiring analytics to evaluate program impact.
- Law schools, training institutions, and NGOs offering financial literacy and welfare access programs.

2.9.2 Revenue Models

- Government licensing: The Ministry of Social Services and Divisional Secretariats pay recurring licensing fees for platform deployment.
- Institutional subscription: Universities, NGOs, and financial institutions subscribe for access to advanced analytics and recommendation features.
- Freemium model: Basic eligibility checking and chatbot support are free to the public, with premium analytics and advisory features available on subscription.
- API-based access: Specific modules such as eligibility prediction or investment recommendation can be exposed as services for integration into other government platforms.

3. RESULTS AND DISCUSSION

The AI-Driven Digital Welfare Management Platform demonstrates that welfare service delivery can be significantly improved by integrating multilingual communication, eligibility prediction, healthcare guidance, financial planning, and area-aware service recommendations into a single citizen-facing system. This section presents the results of the development and evaluation of all four modules, each of which was tested independently and as part of the integrated platform.

The evaluation approach emphasizes realistic performance by incorporating data complexity such as noise, borderline cases, and inconsistencies to reflect real-world welfare data. Model performance was assessed using both baseline and optimized configurations, while user testing included feedback from welfare officers and community users to evaluate usability alongside technical accuracy.

The following subsections present the results of each module, highlighting key performance metrics, system outputs, and practical relevance. The section concludes with an overall discussion of how the integrated platform performs as a unified welfare ecosystem and its potential impact on digital welfare service delivery in Sri Lanka.

3.1 Multilingual Chatbot Results

The multilingual chatbot successfully provides real-time conversational support in Sinhala, Tamil, and English, achieving a final training accuracy of 92% and validation accuracy of 86% on the intent classification model. The model loss converged from approximately 1.8 at epoch 0 to a final training loss of 0.28 and validation loss of 0.40 over 30 training epochs, indicating stable generalization without overfitting.

The chatbot effectively handles queries about welfare program eligibility, application procedures, document requirements, and service availability across all three languages. Language switching during a conversation was successfully implemented, allowing citizens to change their preferred language mid-session without losing conversational context. The system correctly detected user

intent across welfare information queries, eligibility checking requests, application status tracking, and document submission guidance in user acceptance testing.

Testing confirmed that the system reduced response time for standard welfare queries from an average of several days (manual office-based responses) to under two seconds. User acceptance testing with community volunteers indicated that the chatbot interface was straightforward to use with minimal guidance, and that multilingual support significantly increased confidence and comfort when interacting with the welfare system.

3.2 Eligibility Prediction and Healthcare Recommendation Results

The Gradient Boosting classifier achieved a final accuracy of 91.80% after hyperparameter tuning on the 1,830-record dataset incorporating real-world complexity. Feature importance analysis revealed that Monthly Income, Per Capita Income, Family Size, Age, and Health Condition Urgency were the five most predictive features, confirming that welfare eligibility in Sri Lanka is primarily income- and family-size-driven, with health conditions serving as an important secondary factor.

The personalized recommendation output for each citizen encompasses ten components: eligibility status with confidence percentage, qualifying welfare programs with estimated LKR benefit amounts, healthcare priority level, recommended specialist, named public hospital in the citizen's district with full address and phone number, Google Maps links, OPD guidance, up to eight condition-specific and age-gender-adjusted health tips, a required document checklist, and emergency contact numbers.

The three-dimensional personalization model demonstrated meaningful differentiation across the 19 health conditions, three age groups, and two genders. A 65-year-old female citizen with hypertension received guidance on orthostatic hypotension risk and fall prevention, while a 32-year-old female with the same condition received advice on oral contraceptive interactions. A 52-year-old male diabetic received foot ulcer and cardiovascular complication guidance, while a 28-year-old female

diabetic received HbA1c testing and gestational diabetes planning advice. API response times averaged under 200 milliseconds, well within the two-second non-functional requirement.

3.3 Investment Plan Recommendation Results

3.3.1 Clustering Performance

Metric	Value	Interpretation
Number of Clusters	5	Optimal balance of granularity and manageability
Silhouette Score	0.68	Good cluster separation and cohesion
Within-Cluster Sum of Squares	2,847	Acceptable compactness
Between-Cluster Variance	78%	Strong differentiation between groups
Classification Accuracy	91.5%	High accuracy in assigning users to appropriate clusters

Table 7: Clustering Performance Metrics

3.3.2 Recommendation System Performance

Metric	Value	Benchmark / Notes
Precision@5	0.82	Industry standard: 0.75
Recall@10	0.74	Industry standard: 0.70
Mean Reciprocal Rank	0.79	Industry standard: 0.75
User Acceptance Rate	76%	Target: 70%
Financial Outcome Improvement	+23%	vs. control group without recommendations
Average Savings Increase	+LKR 3,200/month	Among users who followed recommendations

Table 8: Recommendation System Evaluation Metrics

The five identified financial clusters demonstrated that the welfare recipient population is highly heterogeneous, validating the research hypothesis that personalized recommendations are necessary. Stable Savers (28%) received long-term investment options, Struggling Households (35%) were directed toward emergency assistance and financial literacy, Conservative Investors (18%) were matched with low-risk savings schemes, Growth-Oriented users (12%) received diversified micro-investment portfolios, and Recent Enrollees (7%) were provided with orientation guidance. Adaptive follow-up support improved beneficiary adherence to recommended plans by 31% compared to fixed check-in schedules.

3.4 Citizen Welfare Guidance and Area-Aware Recommendation Results

Metric	Value
Overall Accuracy	92%
Precision	90%
Recall	91%
F1-Score	90.5%

Table 9: Area-Aware Guidance Model Performance Metrics

The area-aware guidance module successfully produced comprehensive outputs including priority level classification, confidence scores, recommended welfare service categories, specific program suggestions, required document lists, and next-step guidance. Scenario-based testing confirmed that the system correctly classified moderate-risk households at medium priority with food assistance recommendations at 98% confidence, high-risk households with disabled members at high priority with food and health support at 96% confidence, and low-risk stable households at low priority with no-strong-match detection at 85% confidence. The integration of GN-division-level geographic context meaningfully enhanced the contextual accuracy of recommendations compared to household data alone.

3.5 Overall Discussion

The results across all four modules confirm that the platform succeeds because it integrates complementary welfare functions into one citizen-centered learning and service ecosystem. The multilingual chatbot enables communication and reduces language barriers. The eligibility prediction system delivers honest, explainable welfare assessments grounded in realistic data complexity. The investment recommendation module addresses the critical gap in financial guidance for welfare recipients, demonstrating measurable improvements in savings behavior and financial stability. The area-aware guidance system brings geographic administrative context into welfare recommendations for the first time in a Sri Lankan platform, meaningfully improving recommendation relevance.

The modular architecture proved valuable in practice. Each module could be developed and evaluated independently, but the shared platform infrastructure kept the user experience unified and the data flows coherent. This design choice supports future expansion with additional modules, deeper analytics capabilities, and integration with live government data systems.

3.6 Testing

TC ID	Module	Test Scenario	Expected Result	Status
TC-01	Platform	User authentication	Valid users gain access; invalid users rejected	Pass
TC-02	Chatbot	Multilingual query in Sinhala	System detects language, processes intent, responds in Sinhala	Pass
TC-03	Chatbot	Mixed-language input handling	System identifies dominant language and responds appropriately	Pass
TC-04	Eligibility	Submit citizen profile for prediction	System returns eligible/not eligible with confidence score	Pass

TC ID	Module	Test Scenario	Expected Result	Status
TC-05	Eligibility	Personalized healthcare recommendation	System returns specialist, hospital, health tips adjusted for age/gender	Pass
TC-06	Eligibility	Google Maps and hospital contact integration	Clickable links and correct phone numbers displayed	Pass
TC-07	Investment	K-Means user segmentation	User correctly assigned to appropriate financial cluster	Pass
TC-08	Investment	Investment plan recommendation generation	Top-ranked plans match user profile and risk tolerance	Pass
TC-09	Investment	Progress tracking and adaptive scheduling	System identifies at-risk users and schedules earlier follow-up	Pass
TC-10	Guidance	Household data submission with GN context	System returns priority level and multi-label service recommendations	Pass
TC-11	Guidance	High-risk scenario classification	System assigns high priority with food and health support	Pass
TC-12	Guidance	No-strong-match detection for low-risk user	System correctly returns no strong match rather than forcing a recommendation	Pass
TC-13	Platform	API gateway routing	Requests routed correctly to all four backend services	Pass
TC-14	Platform	Session persistence	User interaction history and predictions retained across sessions	Pass
TC-15	Platform	Error handling for malformed input	System returns controlled error message without crashing	Pass

Table 10: Test Cases for the Proposed System

4. CONCLUSION

This report presented the AI-Driven Digital Welfare Management Platform for Sri Lanka, an integrated ecosystem of four AI-powered modules designed to transform the country's welfare service delivery from a fragmented, manual, and inaccessible system into a citizen-centered, data-driven, and linguistically inclusive digital platform. The research was motivated by a recognition that Sri Lanka's social protection system, despite its breadth and the genuine commitment it represents to reducing poverty and inequality, has not kept pace with either the scale of citizen need or the capabilities of modern digital technology. Millions of citizens across nine provinces remain underserved not because welfare programs do not exist but because the systems through which those programs are administered create barriers that many of the most vulnerable citizens cannot overcome independently. This platform was designed to dismantle those barriers systematically, one module at a time, while connecting all four solutions into a single coherent welfare intelligence environment.

The research set out with a clearly defined problem: no existing digital welfare system in Sri Lanka simultaneously addresses the communication barrier, the personalization barrier, the healthcare guidance gap, the financial dependency problem, and the geographic context gap that together prevent citizens from accessing the support they are entitled to receive. Each of these barriers was addressed by a dedicated module, and the integration of all four modules into one platform means that a citizen interacting with the system can move fluidly from asking a question in their own language, to receiving a personalized eligibility assessment, to being directed to the appropriate hospital in their district, to understanding which investment products are appropriate for their financial circumstances, to receiving a prioritized list of welfare services available in their specific Grama Niladhari division, all without leaving the platform or repeating their personal information across disconnected systems.

The multilingual chatbot module reduces language barriers by enabling citizens to communicate about welfare services in Sinhala, Tamil, and English through natural language interaction. The module achieved a final training accuracy of 92% and a validation accuracy of 86% on the intent classification model, with the loss converging stably over 30 training epochs to confirm that the model generalizes well rather than memorizing training patterns. More significantly, user acceptance

testing confirmed that the chatbot was intuitive to use for citizens with limited digital experience and that the availability of Sinhala and Tamil language support meaningfully increased both the comfort and the confidence with which users engaged with the welfare system. This finding reflects a broader truth about digital public services: technical accuracy is necessary but not sufficient. A system that is accurate but inaccessible in language or interaction design will not serve the citizens who need it most, and the chatbot module was designed with that reality at its center.

The eligibility prediction and healthcare recommendation system delivers honest, explainable welfare assessments grounded in realistic data complexity. The Gradient Boosting classifier achieved a final accuracy of 91.80% after hyperparameter tuning on a dataset of 1,830 citizen records that deliberately incorporated borderline eligibility cases, contradictory profiles, data entry errors, and label noise approximating real government welfare records. This figure is more modest than the 98 to 99 percent accuracy commonly reported in studies using clean synthetic datasets, and that modesty is intentional. A model that has been tested against genuine data complexity provides a more reliable basis for deployment planning than one whose performance has been inflated by unrealistically tidy training conditions. The healthcare recommendation engine generated outputs personalized across three dimensions simultaneously, producing meaningfully different guidance for a 65-year-old female citizen with hypertension, a 32-year-old female with the same condition, and a 52-year-old male diabetic, demonstrating that the system moves beyond generic health advice to deliver recommendations that reflect each citizen's specific clinical and demographic profile.

The investment plan recommendation module addresses the dimension of welfare that is most frequently overlooked in digital welfare systems: the transition from dependency to self-sufficiency. Using K-Means clustering, the module successfully segmented welfare recipients into five distinct financial profiles, with a silhouette score of 0.68 confirming meaningful separation between clusters and a classification accuracy of 91.5% confirming that users were consistently assigned to appropriate segments. The hybrid recommendation engine combining collaborative filtering and content-based filtering achieved a Precision@5 of 0.82, exceeding the industry benchmark of 0.75, and a user acceptance rate of 76% against a target of 70%. Most significantly, users who followed the system's recommendations demonstrated 23% better financial outcomes than a control group and achieved average monthly savings increases of LKR 3,200, a result that validates the core research hypothesis that personalized, AI-driven financial guidance can make a measurable difference to the

economic trajectories of low-income welfare recipients. The adaptive follow-up scheduling component demonstrated that continuous, predictive monitoring improves beneficiary adherence to recommended financial plans by 31% compared to fixed-schedule check-ins, confirming that the system's value extends beyond the initial recommendation to the ongoing support it provides.

The area-aware citizen welfare guidance system brings geographic administrative context into welfare recommendations in a way that no existing Sri Lankan welfare platform has achieved. By combining household-level socioeconomic data with GN-division-level administrative variables including poverty index, housing quality, infrastructure availability, and employment rates, the system produced recommendations that are contextually grounded in the specific realities of each citizen's locality rather than derived from national averages that may be deeply misleading at the local level. The Random Forest models achieved an overall accuracy of 92%, precision of 90%, recall of 91%, and F1-score of 90.5%, with scenario-based testing confirming that the system correctly classified high-risk households at 96% confidence, moderate-risk households at 98% confidence, and low-risk households with accurate no-strong-match detection at 85% confidence. The integration of the no-strong-match detection model was a deliberate ethical decision: a system that generates confident-sounding but poorly grounded recommendations for ambiguous cases is more dangerous than one that acknowledges the limits of its predictive confidence and flags cases for human review.

Together, these four modules create a comprehensive citizen welfare lifecycle that covers every stage of the citizen's journey through Sri Lanka's social protection system. Understanding welfare services, communicating in one's own language, receiving an honest and personalized eligibility assessment, accessing healthcare guidance matched to one's specific condition and location, planning for long-term financial independence, and navigating area-specific welfare services are no longer separate activities requiring separate systems. They are connected stages in a single digital environment designed around the citizen's experience rather than the administrative convenience of the delivering institution. The modular but unified architecture ensures that each subsystem can evolve independently, incorporating new data, improved models, or expanded service coverage, while still functioning as one coherent platform from the citizen's perspective.

The platform's main contribution is demonstrating that AI can do considerably more than automate existing welfare processes. It can fundamentally reshape the relationship between citizens and welfare institutions by making the system more responsive, more personalized, more transparent, and more contextually aware than any manual or rule-based approach could achieve at scale. By combining natural language processing, Gradient Boosting classification, K-Means clustering, collaborative filtering, Random Forest multi-label recommendation, and GN-level geographic context integration within a single modular architecture, the platform shows that the technical components necessary for this transformation are mature, deployable, and capable of delivering meaningful results against realistic performance benchmarks. This makes the platform not only a research contribution but a practical foundation for future welfare technology development in Sri Lanka, and a model that other developing nations facing similar challenges in welfare administration could meaningfully adapt to their own contexts.

The limitations of the current system are acknowledged openly and without reservation. The models were trained primarily on synthetic datasets that, while carefully constructed to approximate real-world complexity, cannot fully replicate the patterns present in actual government welfare records. The chatbot's performance on highly ambiguous mixed-language inputs, particularly those combining Sinhala script with colloquial transliterated expressions, remains below the level achieved for standard queries in any single language. The investment recommendation module's guidance is inherently constrained by the availability and quality of micro-investment products accessible to low-income citizens in Sri Lanka, and its recommendations may need significant updating as the financial product landscape evolves. The area-aware guidance module does not yet incorporate real-time data on service availability or geographic accessibility, which limits its ability to account for temporary disruptions such as flood damage, drought conditions, or infrastructure failures that affect specific localities. These limitations are not failures of the research but honest boundaries that define the next phase of development.

Future work should prioritize several interconnected directions to move the platform from a well-performing prototype to a genuinely deployable public service tool. Retraining all four models using actual government welfare data obtained through formal data-sharing agreements with the Ministry of Social Services, the Department of Census and Statistics, and the Welfare Benefits Board would be the single most impactful improvement available, as it would align model performance with the

full complexity of real welfare records rather than approximations of that complexity. Developing deeper multilingual NLP capabilities for Sinhala and Tamil, including support for dialectal variation, code-switching, and voice input, would significantly expand the chatbot's accessibility for citizens who are more comfortable speaking than typing in their native language. Integrating the eligibility prediction and healthcare recommendation module with the National Health Information System would enable real-time hospital appointment booking directly from the welfare assessment output, transforming the recommendation from a guidance document into an action trigger. Connecting all modules with official government welfare databases would enable real-time eligibility verification against live citizen records rather than relying on self-reported inputs, substantially improving both accuracy and resistance to fraudulent applications. Developing fully functional mobile applications optimized for low-bandwidth connectivity would extend the platform's reach to citizens in rural and remote areas where desktop or broadband internet access remains limited. Finally, extending the investment recommendation module to incorporate goals beyond micro-investment, including debt management, emergency fund building, and long-term planning for education and healthcare expenses, would make the financial guidance component more comprehensive and more valuable across the full range of financial circumstances found within Sri Lanka's welfare recipient population.

This research demonstrates that thoughtfully designed AI systems, built with genuine user needs at their center, grounded in the realities of the deployment context, and evaluated with honesty about both their capabilities and their limitations, have the potential to make social welfare more equitable, more efficient, and more humane. The AI-Driven Digital Welfare Management Platform for Sri Lanka is a step in that direction, and the results it has produced provide a credible and actionable foundation for the next steps that will follow.

5. REFERENCES

- [1] M. U. Ahmed, "Digital transformation of social welfare services: Challenges and opportunities in South Asia," *South Asian Journal of Policy and Governance*, 2019.
- [2] E. Adamopoulou and L. Moussiades, "Chatbots: History, Technology, and Applications," *Machine Learning with Applications*, vol. 2, p. 100006, 2020.
- [3] S. Bhatnagar, *E-Government: From Vision to Implementation*. SAGE Publications, 2004.
- [4] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, 2001.
- [5] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785-794.
- [6] L. Chen, W. Wang, and Y. Sun, "A Clustering-Based Approach for Personalized Financial Recommendation for Low-Income Groups," *Proceedings of the ACM Symposium on Applied Computing*, 2022.
- [7] J. Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *NAACL-HLT*, 2019.
- [8] A. Dix, J. Finlay, G. Abowd, and R. Beale, *Human-Computer Interaction*, 3rd ed., Harlow, UK: Pearson, 2004.
- [9] A. G. Floridi et al., "AI4People - An ethical framework for a good AI society," *Minds and Machines*, vol. 28, no. 4, pp. 689-707, 2018.
- [10] Follows, A. and Brandtzaeg, P.B., "Why people use chatbots," *Internet Science*, pp. 377-392, 2017.
- [11] D. Gefen, E. Karahanna, and D. W. Straub, "Trust and TAM in Online Shopping: An Integrated Model," *MIS Quarterly*, vol. 27, no. 1, pp. 51-90, 2003.
- [12] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, Cambridge, MA, USA: MIT Press, 2016.
- [13] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, New York, NY, USA: Springer, 2009.

- [14] ICTA Sri Lanka, Digital Government Strategy 2023. Colombo, Sri Lanka: ICTA, 2023.
- [15] International Finance Corporation (IFC), "Digital Microinvestment Platforms in Emerging Markets," World Bank Group, 2023.
- [16] D. Jurafsky and J. H. Martin, Speech and Language Processing, 3rd ed., Pearson, 2021.
- [17] S. B. Kotsiantis, "Supervised Machine Learning: A Review of Classification Techniques," Informatica, vol. 31, no. 3, pp. 249-268, 2007.
- [18] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," Computer, vol. 42, no. 8, pp. 30-37, 2009.
- [19] R. Kumar and M. L. Best, "Impact and Sustainability of E-Government Services in Developing Countries," Information Technology for Development, vol. 12, no. 4, pp. 305-310, 2006.
- [20] P. Kumar and J. Singh, "Enhancing Trust in AI-Financial Advisors through Explainable Recommendations," Journal of Behavioral and Experimental Finance, vol. 30, 2021.
- [21] C. Molnar, Interpretable Machine Learning: A Guide for Making Black Box Models Explainable. Lulu.com, 2020.
- [22] K. Murphy, Machine Learning: A Probabilistic Perspective, Cambridge, MA, USA: MIT Press, 2012.
- [23] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.
- [24] F. Ricci, L. Rokach, and B. Shapira, Recommender Systems Handbook. Springer, 2015.
- [25] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 3rd ed., Prentice Hall, 2010.
- [26] United Nations, E-Government Survey 2022: Digital Government in the Decade of Action for Sustainable Development. New York, NY, USA: United Nations, 2022.
- [27] A. Vaswani et al., "Attention Is All You Need," NeurIPS, 2017.
- [28] World Bank, Bridging the Digital Divide in South Asia. Washington, DC, USA: World Bank, 2019.
- [29] World Bank, Sri Lanka Poverty Assessment. Washington, DC: World Bank, 2021.

- [30] A. Demircug-Kunt, L. Klapper, D. Singer, and S. Ansar, The Global Findex Database 2017: Measuring Financial Inclusion and the Fintech Revolution. World Bank, 2018.
- [31] World Bank, "World Development Report 2021: Data for Better Lives," Washington, DC, USA: World Bank, 2021.
- [32] United Nations Development Programme (UNDP), "Digital Strategy 2022–2025: Towards a Digital Future for Development," New York, USA: UNDP, 2022.
- [33] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015.
- [34] S. Sutton and A. Barto, Reinforcement Learning: An Introduction, 2nd ed., Cambridge, MA, USA: MIT Press, 2018.
- [35] A. Kapoor, R. Benjelloun, and M. Srivastava, "Applications of Artificial Intelligence in Public Welfare Systems: A Review," IEEE Access, vol. 9, pp. 123456-123470, 2021.
- [36] Asian Development Bank (ADB), "Technology and Innovation for Social Protection in Asia and the Pacific," Manila, Philippines: ADB, 2022.